Text Summarization of Indian Languages: A Review

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Abstract

Automatic Text Summarization (ATS) is becoming much more important because of the huge amount of textual content that grows exponentially on the Internet and the various archives of news articles, scientific papers, legal documents, etc. Text summarization automatically produces a summary containing important sentences and includes all relevant important information from the original document. One of the main approaches, when viewed from the summary results, is extractive and abstractive. This paper hands out a broad overview on the work done in the field of automatic text summarization in Indian languages using various text summarization approaches. The focal centre of this review paper is to present the research done on text summarization on languages such as, Hindi, Punjabi, Bengali, Malayalam, Kannada, Marathi, Assamese, Konkani, Odia, and Telugu. The paper also discusses the evaluation techniques being used for assessing the summaries.

Keywords Text Summarization |Natural Language Processing| Approaches| Low resource Language

1 Introduction

Today a large number of data is being produced online from websites, news, social media, etc. In this busy world, data is needed to be concise for the people. Automatic Text Summarization the new state-of-the the art is used to summarize the data into relevant and precise form. Text summarization has been applied in various fields such as news summary, legal documents, financial marketing, social media marketing, biomedical, etc[1][5]. Text summarization is broadly categorized into 2 approaches: extractive , abstractive. In extractive text summarization the summary is made from the relevant sentences in the document[2]. A lot of work research and work has been done in extractive summarization as compared to abstractive text summarization. The various methods used in extractive summarization are statistical-based, graph-based, deep learning-based, semantic-based, topic based, etc. In abstractive text summarization the summary of the text is created by the model by paraphrasing using new words. There is still a lot of work going in the abstractive text summarization [3][4]. In NLP a lot of work has been done in text summarization, especially for high level languages such as English, Chinese, Arabic etc. But a lot of work is still going on in low resource languages. There are very few papers on the text summarization of Indian languages approaches. In this paper we will review about the low resource Indian languages such as Hindi, Punjabi, Malayalam,Tamil, Urdu, Marathi etc. and various approaches and techniques that have been used in these Indian languages for the automatic text summarization.

2 Different Techniques Used in Text Summarization

The different methods used for different languages for the summarization have been broadly categorized into 5 techniques.

1.Machine learning- The machine learning-based approach learns from the data. It can be supervised or unsupervised. In supervised approach, there is a collection of documents and their respective human-generated summaries such that useful features of sentences can be learned from them. The system classifies each sentence of the test document either into “summary” or “non-summary” class with the help of a training set of documents. Whereas in unsupervised learning no such training data is required. Fuzzy, SVM, and Naïve Bayesian methods are some of the supervised approaches that have been used for the summarization. The clustering and K-means are examples of unsupervised machine learning.[2][28]

2.Deep Learning – The deep learning technique comes from the machine learning approach only. It has majorly helped in developing abstractive summarization models. Methods like Neural networks, RNNs, LSTM, Seq2Seq etc. showed promising results for the short text summarization.

However, they still suffer from the inability to deal with out-of-vocabulary (OOV) words and also unable to generate repeated words or phrases.[28]

3.Rule-based- In this approach the methods are required to define the rules and categories to discover the important aspects of the input text to generate the summary. Most of the rules are created manually according to the requirement. The rules are defined on methods like Topic based, cue phrases, word frequency, Title overlap etc. These rule based have high information density and are able to expand variety of the abstraction schemes and generation patterns to handle more aspects. The main disadvantage of this is it is very tedious work and all the rules have to be written manually.[2]

4.Graph-based- In this the sentences are represented as nodes and edges connect the related elements together. The sentences are selected by comparing and if they lie above the given limit then a connection is setup between them in the graph. This system is called the LexRank. The mathematical part is known as the cosine similarity. The methods used in this techniques are semantic graph, Minimum spanning Tree, thematic relation based graph approach. The limitation of this technique is that they do not reflect the word/phrase meaning.[2][28]

5.Statistical – This technique is very useful for text summarization and has been used for many languages. It does not require any additional linguistic knowledge. It also helps in reducing redundancy and can be used with any of the other techniques. Some of the statistical features are the position of the sentence, positive keyword (based on frequency count), negative keyword (based on frequency count), the centrality of a sentence (i.e. similarity with other sentences), the resemblance of the sentence to the title, relative length of the sentence, presence of numerical data in the sentence, presence of proper noun, etc. For each sentence in the document score is calculated and based on scores the highly scored sentences are chosen for generating the summary. The other methods used are TF-IDF, sentence weight learning, and Singular Value Decomposition(SVD). The limitation of the statistical technique is that the summary generated is not well structured. [28]

3 Low Resource Indian Languages

Hindi

Hindi is the official language of India. It is spoken vastly in the northern part of India.Vijay used statistical and linguistic features approach for the extractive text summarization of hindi. They took a dataset of 24253 articles summary of an online newspaper and evaluated the summary on couple of parameters. Three main steps were used to summarize the text: pre-processing, processing, post-processing. They found out that the recall score was 70% and the precision was 62% [6]. Further Gulati&Sawarkar proposed a system for multi-document Hindi summarization. They used fuzzy logic classifiers and eleven features. The major steps involved were: pre-processing, extraction of feature terms and fuzzy logic for ranking the sentence based on the optimized feature weights. News articles from Hindi newspaper were used as dataset. The experimental results showed an average precision of 70%[7]. Manish et al in their research described an approach in which 30 hindi documents were tested and applied a rule based approach which defines 18 features. Handcrafted rules are developed to create the summary of the text documents written in Hindi language. The overall accuracy is achieved to be 96% which is better than many techniques[8]. Then Dalal&Malik used semantic graph based approach for automatic text summarization using the Particle Swarm Optimization (PSO) which is very powerful technique for optimal solution. It is a bio-inspired method. After which the trained classifier are used to derive sub-graph form the semantic graph to generate the summary. 80 hindi documents were used as the dataset. A result with recall of 60%, precision 42.86%, F-score of 50.01% was found out[9].

Punjabi

Punjabi is a low resource language which is mostly spoken in Punjab and some parts in north of India.Gupta et al were the first to develop an online news text summarizer in Punjabi application using the sentence weight learning technique. The technique has 2 major steps: processing and pre-processing. A dataset of 50 news documents was experimented which contained various articles in Punjabi. The experimental research gave F-score of 95.32% and an accuracy of 81.38%[10]. Kaur et al combine statistical and linguistic approach for the extractive text summarization. The second method they used was the SVM. The research found out that the SVM is better than many machine learning techniques. They used 150 news documents and Unicode based Punjabi corpus. The average precision was 82% and the average recall was 81%. The ROUGE-2 score was 85%.[11]. Further Jain et al proposed a Neural network technique and the dataset was taken from ILCI. The method has three major steps: pre-processing, processing and classification. In classification neural networks is used with sigmoid as the activation function. The precision, recall and F-measure are achieved as 90.0%, 89.28% an 89.65% respectively which is reasonably good in comparison to the performance of other existing Indian languages summarizers. [12]

Kannada

Kannada is one of the most ancient Dravidian language which is approximately spoken by 40 million people. Geetha et al experimented with the Latent Semantic Analysis which is an analytic tool for the extractive summarization of Kannada. They used the kannada Wikipedia as the dataset for the experiment. The LSA extracts the source text and converts it into the term sentence matrix using the Singular Vector Decomposition (SVD). SVD is used to find the correlation between the words and sentences. Parameters like, precision and accuracy were evaluated. The results were evaluated with gold summary standards. It showed an accuracy of 94% and precision 80% [13]. It is seen that LSA performance decreases with large documents and multilingual text.

Malayalam

In 2014 Kabeer et al proposed two methods that are statistical sentence scoring technique and semantic graph based method for extractive and abstractive text summarization respectively[14].The data set used for evaluation consist of 25 Malayalam documents from the news website and their human written summaries. The sentence scoring technique has 3 main steps : pre-processing, keyword identification and summary extraction. Further the graph based technique has 5 major steps: Semantic Triple Extraction, Anaphora resolution, semantic graph generation, semantic graph reduction and summary generation. The performance was evaluated both manually and automatically. In both the performance, result of sentence scoring was a little better than the semantic graph method. The ROUGE-L score for Statistical method was 0.637 precision, 0.0.65 recall and 0.533 F-measure. Whereas for the semantic method the precision, recall and F-measure scores were 0.466, 0.667 and 0.400 respectively. Then (Ajmal et al) proposed a graph theoretic approach for extractive text summarization. In this two sentences are connected with an edge and their cosine similarity is greater than the threshold thus helps in identification of important sentences in the document which are preferred to be included in the summary. 6 different datasets were used for Malayalam articles and olam. The various parameters such as precision, recall and F-measure is calculated on those datasets. The average respective scores were 0.64, 0.71, and 0.68 [15]. Further Raj and Haroon proposed a method of minimum spanning tree based Graph based reduction. The main feature of this approach is that MST reduces the graph into sub-graphs which gives a distinct output and error-free approach. The evaluation is done with four different datasets which are selected from the online Malayalam news website and Olam dictionary. The approach gives an average precision of 0.572 and an average recall value of 0.683 and F-measure of 0.587[16].

Nambiar et al proposed an attention-based seq2seq model for generating an abstractive summary. They compared the attention model with the baseline seq2seq model. The attention model consists of encoder, decoder and attention layer. The BBC news dataset was trained for single document summarization. The results showed that the baseline seq2seq performs poorer than the attention learning model[17]. The ROUGE-L score for attention is 27.41 thus showing promising results.

Marathi

Very little work has been done in developing a Marathi text summarizer. Sarwadnya and Sonawane proposed a Marathi extractive text summarizer using graph based model. A dataset of 634 documents based on Marathi news articles were extracted. The system was broadly divided into two main steps: pre-processing and processing. In pre-processing stop word removal, lemmatization and stemming were applied. Whereas in pre-processing feature extraction and text rank algorithm was performed. At last the summary is generated through summary generation. The ROUGE-n and F-measure were evaluated. The result shows good scores for both the parameters as well as method. It is seen that TextRank when used along with thematic similarity gives best precision, while TextRank with positional distribution gives best recall. The score for ROUGE-1 till 4 is 0.86, 0.86, 0.85, 0.84 [18].

Bengali

Uddin and Khan in 2007 proposed a text ranking method. Various features such as term frequency, cue phrases, Title etc. were used for generating the summary. Bengali newspaper text was used as the dataset for the experiment. They evaluated the result manually by evaluating 10 different summarized articles. It gave an average accuracy of 84%[19]. Then Das and Bandyopadhaya proposed an opinion based summarization system. They developed a Document Level Theme Relational Graph which is used for generating the final summary. A Bengali online news corpus has been used as the dataset. The features were broadly categorized into 3 i.e. Lexico-Syntactic, Syntactic and Discourse level features. The summarization system has been evaluated with Precision of 72.15%, Recall of 67.32% and F-measure of 69.65% [20].

Odiya

Biswas & Acharya et al experiment with various methods for the Odiya text summarization. The methods used are the word frequency method, Positional Criteria method, Cue phrase method, Title overlap method. Precision and recall are evaluated. In Word frequency, the summary is extracted using the words with higher frequency. It gives a precision

and recall of 83%. In positional criteria, the first and last sentences are extracted. Also, sentences with features like bold, italics, and underline are also extracted. It gives a precision and recall of 100%. The cue phrase method is based on the assumption that phrases like “in conclusion” and “note that” provide a context for identifying important sentences for the summary. They identified 37 cue words. The precision and recall was 100%. The title overlap method morphological analyzer is used for better accuracy. It also gives the same result as previous methods. Thus, all 4 methods show promising results with accuracy of approximate 95% [21]. Nayak et al used the cosine similarity and clustering approach for the Odiya text summarization. The news documents have been used as the dataset for the method. After the pre-processing the cosine similarity is applied among the sentences and a matrix is generated which tells about the relevance of the text in the document. In the final step, the agglomerative hierarchial clustering is applied. The F-score is used as a metric for the evaluation of the proposed system and gives a satisfactory result of 60.27% [22].

Assamese

In 2020 Baruah et al proposed a text summarization of assamese by incorporating linguistic and statistical features. They developed a modified tf-idf also. The approach has 3 main steps: Pre-processing, sentence scoring and summary generation. In the sentence scoring various features are selected for scoring of sentences. They used a modified tf-idf rather than the traditional one. In the summary generation the overall all score is calculated and the sentences with higher score are extracted. They used a dataset of 20 lakh Assamese words which was categorized into cultural, religion and international news. The summaries extracted were compared to the human generated summaries by applying ROUGE-1 and ROUGE-2 to find the precision, recall and f- measure. The avg. precision with the CR 50% for ROUGE-1 and ROUGE-2 0.59 and .50 respectively. The average recall was 0.68 and 0.57. The f- measure was 0.63 and 0.53 [23]. The Assamese language is still lagging behind compared to other Indian languages due to limitation of resources.

Telugu

Telugu is the third largest speaking language in India. Still there are very less resources and very less work has been done on the text summarization of telugu. Nagaprasad et al proposed a SVM method for the text summarization. The 300 telugu news articles are used as the dataset. The f1 score was evaluated for the performance. The f1-score was 0.82 which is a good score. More features can be added in the method for better results[24]. Then Sudha and Latha used deep learning technique for the summarization. The RNN technique is used in this deep learning approach. The heterogenous multi- documents are used as the dataset. The precision, recall and f-score are evaluated for the performance of the approach. The precision, recall and f-score are 0.016, 0.041 and 0.029 respectively[25].

Konkani

Rodrigues & Fernandes et al proposed the sentence extraction method for the extractive text summarization of the Konkani language. The method has three major steps. First pre-processing of the dataset is done which includes tokenization, stop word removal and segmentation. Then the sentences are ranked based on the following features: sentence position, keyword feature and sentence length. At last the sentences are arranged in descending order and the highest ranked sentences are extracted. For the experiment Konkani newspaper Bhaangarbhuin has been used as the dataset. The summarization methods in this study was evaluated automatically using the text summarization evaluation toolkit, ROUGE. The performance evaluation of the three documents were the following-: Doc1 recall 0.3196, precision 0.3939 and f-score 0.35. Doc2 recall is 0.5263, precision 0.6000 and f-score 0.56. For Doc3 recall is 0.25, precision is 0.15 and f-score 0.18[26]. The relevance of the summary with respect to the human analysis was found to be 46.89%. Further D’Silva and Sharma proposed an Automatic text summarization of Konkani texts using pre-trained word embedding and deep learning. The FastText pre-trained is used to generate word embedding and then the multi-layer perceptron MLP is used for the text summarization. The dataset comprises of 71 folk tales from the Konkani literature. The method has 4 major steps: pre-processing, compute sentence, Classification with K-fold cross validation and summary generation. The ROUGE-L score of fast text Deep MLP is closed to the human-annotate benchmark. The precision, recall and F-score is 0.329,0.320,0.324 respectively [27].

Languages	Author	Technique	Dataset	Performance parameter
Hindi	Dalal&Malik	Data clustering approach	80 hindi docs	Recall=60% Precision=42.86% F1-score=50.01%
	Vijay	Statistical& linguistic	Online news	Recall=70% Precision=62%
	Gulati and Sawarkar	Fuzzy logic classifier&eleven features	News articles from Hindi newspaper	Average Precision=70%
	Manish&Naresh	Rule based approach	30 Hindi documents	Accuracy=96%
Punjabi	Gupta& Lehal	Sentence Weight learning	50 news documents	F-score=95.32% Accuracy=81.38%
	Gupta& Kaur	Statistical &linguistic	1.150 news documents 2. Unicode based Punjabi Corpus	Average precision=82% Avg. recall=81% Avg. ROUGE-2= 85%
	Jain& Arora	Neural Network	Monolingual Punjabi Text Corpus	Precision=90% Recall=89.28% F-score=89.65%
	Geetha	Singular Value Decomposition	Arts,Commerce,Homopathy Literature.	Accuracy=94% Precision=80%
Malayalam	Kabeer&Suman	Statistical Sentence Scoring & Semantic Graph	25 Malayalam docs from Malayalam news	Statistical Precision=0.485 Recall=0.525 F-measure=0.590 Semantic Precision=0.455 Recall=0.4968 F-measure=0.4623
	Ajmal & Rosna	Graph Theoretic Approach	Online Malayalam articles & olam dataset	Avg. Precison=0.64 Avg. Recall=0.71 Avg. F-score=0.68
	Raj & Rosna	MST based Graph Reduction	Malayalam news resource	Precision=0.572 Recall=0.683 F-measure=0.587
	Nambiar	Attention based Seq2Seq	BBC news dataset	ROUGE-L =27.41
Marathi	Sarwadnya & Sorawane	Graph based approach	634 news articles	Textrank+Positional F-score= 0.800 TextRank + Similarity F-score=0.8684
Bengali	Das & Bandyopadhaya	K-means clustering	Bengali news Corpus	Precision=72.15% Recall=67.32% F-measure=69.65%
	Udin & Khan	Text ranking method	Bengali newspaper	Accuracy=84%
Odiya	Biswas	1.Word Frequency 2.Positional Criteria 3.Cue Phrase 4.Title Overlap	20 documents odiya dataset	Accuracy Approx=95% Of 4 novel method
	Nayak	Clustering using K-means& cosine similarity	DUC 2003 dataset	F-score=60.27%

Assamese	Baruah	Statistical&linguistic	20 lakh assamese word	50% CR ROUGE-1 Precision=0.59 Recall=0.68 F-measure=0.63 ROUGE-2 Precision=0.50 Recall=0.57 F-measure=0.53
Telugu	Nagaprasad	SVM method	300 telugu news articles	F-score=0.82
	Sudha & Latha	RNN	Heterogenous multi documents	Precision=0.016 Recall=0.041 F-score=0.029
Konkani	Rodrigues & Fernandes	Sentence Extraction	Konkani news	Recall=0.250 Precision=0.152 F-score=0.189
	D'Silva & Sharma	Word embedding and Deep Learning	71 folk tales from Konkani literature	ROUGE-L Recall=0.32 Precision=0.329 F-score=0.324

Table 1 Comparative Analysis of ATS in different Indian languages

4 Parameters

In this section we are going to overview the various parameters that has been used for the evaluation of the generated summary. The metrics used to measure the performance are precision, recall, f-measure and ROGUE.

Precision: It is computed by dividing the number of sentences existing in both reference and candidate (i.e. system) summaries by the number of sentences in the candidate summary. [2]

P=Sref & Scan/Scan

Recall : it is computed by dividing the number of sentences existing in both reference and candidate summaries by the number of sentences in the reference summary. [2]

R= Sref & Scan/Sref

F-measure: It is calculated by using both the precision and recall. It is the harmonic mean between the precision and recall. The mathematical equation is given as following:

F-measure= 2*(Precision)(Recall)/Precision+Recall

ROUGE(Recall Oriented Understudy for Gisting Evaluation) is a metrics which is used for evaluating the automatic text summarization and machine translation. It automatically compares the generated summary or translation with the reference summary or translation. The disadvantage of ROUGE is that it only measures the syntactical matches but is unable to match semantically. There are various ROUGE metrics.[2][5]

ROUGE-1- it is based on the uni-gram measure between a candidate summary and a reference summary.

ROUGE N- is an n-gram recall between a candidate summary and reference summaries.

ROUGE-L – is based on the longest common subsequence (LCS) between our model output and reference, i.e. the longest sequence of words not necessarily consecutive, but still in order that is shared between both.

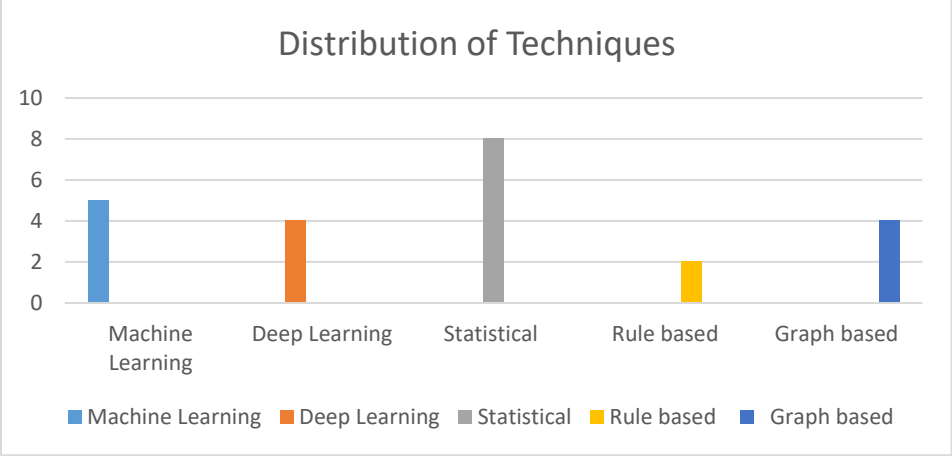

Fig 2 Distribution of Techniques

5 Analysis

There are 22 scheduled languages in India. From the above table and graph, under this section, analysis on the text summarization of the 10 Indian languages is done. It can be concluded that significant work has been done in the field of ATS for the Indian languages with optimal results. But still there is a lot of scope of improvement in the performance as there are not enough resources for such low resource languages. It is found that the most common dataset used for each language are the news articles in online or offline form. The most common technique used for the summarization of these languages is statistical approach. It can be inferred Hindi and Punjabi are the language among all the languages for which the work is done for maximum techniques for the text summarization.

6 Conclusion & Future Scope

The idea of this review paper is to give a brief overview of the research that has been done in the field of automatic text summarization of the 10 most spoken Indian languages i.e. Hindi, Punjabi, Bengali, Marathi, Malayalam, Kannada, Telugu, Assamese, Konkani, and Odiya using various text summarization techniques. It is found from the review that statistical and machine learning techniques are used the most for summarization. It is also interesting to see that majority of research is done for extractive summarization and there are no language in which some research is there on abstractive summarization. Also, one of the important limitation for the Indian languages is that there is not enough dataset. Though the various methods used for the summarization gives a good performance on different parameters like precision, recall, f-score, ROUGE and accuracy but still there is a lot of scope of improvement on the result of these summarized text. In future a lot of research can be expected to be done on abstractive summarization which is considered to show better results than extractive summarization. The availability of dataset for some particular languages is also a novel future aspect in this field. There is a lot of scope in improvement of the results.

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